

SHAUNAK RANE

Software Engineering | Python | Backend Systems | AI/ML

Thane, India | +91 93202 21211 | shaunakrane914@gmail.com | linkedin.com/in/shaunak-rane | github.com/Sh Shaunakrane914 | shaunak-rane.pages.dev

EDUCATION

Universal AI University - B.Tech in Artificial Intelligence & Machine Learning

Expected May 2028 | Karjat, Maharashtra

- **CGPA: 8.47 / 10.0** | **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks, System Design.

TECHNICAL SKILLS

Languages & Core: Python, C, SQL, Bash, Git, Linux (CLI, Scripting)

Backend & Infrastructure: FastAPI, Flask, RESTful APIs, WebSockets, PostgreSQL, Supabase, MySQL, Docker, Linux Internals

Software Engineering: Data Structures & Algorithms (DSA), OOP, pytest (Unit & Integration Testing), Alembic (Schema Migrations), CI/CD

AI & Intelligent Systems: Multi-Agent Workflows, RAG Architecture, PyTorch, PyTorch Geometric, Vector Embeddings (pgvector), Time-Series Modeling

WORK EXPERIENCE

Univitt Technologies - AI & Systems Engineering Intern

May 2026 - Aug 2026 | Remote

- Engineered predictive Condition-Based Monitoring (CBM) pipeline processing 4,000+ daily historian records across 3 industrial plants, reconciling multi-stage thermodynamics.
- Re-implemented historical Degradation Factor Index (DFI) models across 3,879 historian logs with numerical convergence ($<1e-10$ residual vs baseline); formulated deterministic maintenance-reset logic validated across 17 automated regression test gates.
- Containerized Flask REST inference microservices with Docker for edge deployment, delivering sub-second anomaly detection and thermodynamic scoring endpoints.
- Audited 400+ physical and mechanical process variables, evaluating ML models against Holt damped-trend statistical baselines; instituted automated CV promotion gates that halted ungrounded maintenance alerts.

Univitt Technologies - Software Engineering Intern

Apr 2025 - Jul 2025 | Remote

- Developed RESTful backend APIs and normalized relational PostgreSQL database schemas with FastAPI for an institutional food operations and cost-management platform (serving 1,200+ daily meals).
- Automated weekly menu planning and multi-tier Bill-of-Materials (BOM) ingredient rollups, eliminating manual inventory calculation overhead.
- Trained a Random Forest meal-attendance forecaster incorporating weather, calendar events, and shift historical trends, achieving 88% accuracy (within $\pm 10\%$ margin) and outperforming rolling baselines by 14%.
- Maintained 85%+ backend test coverage with pytest, executed Alembic database schema migrations, and collaborated on code reviews using Git pull-request workflows.

PROJECTS

Project Aegis - Threat Intelligence & Claim Verification Engine

Python, FastAPI, Multi-Agent Swarms, RAG, pgvector, Docker

Live Demo: aegis-protocol-110.pages.dev | Source Code: github.com/Sh Shaunakrane914/Misinformation

- Architected decoupled multi-agent verification pipeline (Scout $\rightarrow$ Research $\rightarrow$ Investigator) enforcing strict Pydantic JSON Schema validation, achieving zero unhandled schema errors across benchmark test suites via automated retry policies.
- Engineered quantitative market anomaly detection (Scout Agent) computing 5-day rolling Z-scores on price/volume telemetry via Yahoo Finance proxies to isolate abnormal spikes.
- Implemented canonical SHA-256 claim hashing for sub-millisecond in-memory cache deduplication, preventing redundant LLM token spend, paired with Supabase pgvector semantic retrieval and auditable verdict trails.

TopoFlow - GNN Permeability & Micro-CT Pore Network Benchmark

Python, PyTorch Geometric, GraphSAGE, OpenPNM, FastAPI

Source Code: github.com/Sh Shaunakrane914/Flow

- Benchmarked GraphSAGE against classical Kozeny-Carman physics baselines across 1,231 micro-CT pore networks from 5 geological formations, establishing an empirical regime threshold (C_v) for when graph topology outperforms bulk equations.
- Achieved 46.2% MSE reduction on heterogeneous Savonnières carbonate and 28.4% on Estailades formations over classical solvers, while identifying that homogeneous sandstones remain better predicted by classical physics.
- Built end-to-end pore-network extraction pipeline (PoreSpy/SNOW2), GNN model training workflows in PyTorch Geometric, and streaming FastAPI inference endpoints with SSE.

Gridium Protocol - Autonomous Microgrid Optimization & Telemetry

Python, PyTorch, DDPG, FastAPI, WebSockets, Solidity, Circom

Live Demo: live-ai-1-7tcy.vercel.app | Source Code: github.com/Sh Shaunakrane914/Live-Ai-1

- Developed continuous-control reinforcement learning engine (PyTorch DDPG) in a custom 15-node Ohm's law Gymnasium microgrid (AegisEnv), dynamically adjusting AMM swap fees (0.10%–5.00%) to mitigate solar duck-curve volatility.
- Reduced peak-hour grid strain by 18.4% compared to static flat-fee and rule-based fixed-tariff baselines across simulated multi-node solar generation and battery demand cycles.
- Engineered 500ms Socket.io telemetry gateway, Solidity constant-product AMM contract ($x \times y = k$) with reentrancy protection, and Groth16 zk-SNARK circuits proving energy surplus off-chain.